Exercises 7

Exercise 1 (Pauli matrices and Bell states)**.** *Recall from the lecture notes that the four Pauli matrices are:*

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Also, recall that the four Bell states are:

$$|\beta_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \qquad |\beta_{01}\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle),$$
$$|\beta_{10}\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle), \qquad |\beta_{11}\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle).$$

Finally, recall the vectorization operation:

$$\mathrm{vec}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}.$$

Match each Pauli matrix with the corresponding Bell state. That is, for each Pauli matrix P, find a constant $c \in \mathbb{C}$ and a Bell state $|\beta_{zx}\rangle$ such that

$$\mathrm{vec}(P) = c\sqrt{2}|\beta_{zx}\rangle.$$

Exercise 2 (Local conversion of Bell states)**.** *Verify that any Bell state can be converted to any other by applying some combination of Pauli X and Z matrices to one of the qubits.*

1. *Verify that for any $z, x \in \{0, 1\}$,*
$$|\beta_{zx}\rangle = (Z^z X^x \otimes I)|\beta_{00}\rangle.$$
2. *Verify that for any $z, x \in \{0, 1\}$,*
$$|\beta_{zx}\rangle = (I \otimes X^x Z^z)|\beta_{00}\rangle.$$
3. *Find a sequence S of Pauli X and Z matrices such that*
$$|\beta_{10}\rangle = (S \otimes I)|\beta_{11}\rangle.$$

Exercise 3 (`Cirq`: Superdense coding)**.**

Exercise 4 (SWAP and the Pauli matrices)**.** *Recall that the* SWAP *operation is defined on the computational basis states as follows:*

$$\text{SWAP}\,|a,b\rangle = |b,a\rangle$$

for all $a, b \in \{0,1\}$.

1. *Verify that*
$$\text{SWAP} = \frac{1}{2}(I \otimes I + X \otimes X + Y \otimes Y + Z \otimes Z).$$
2. *Verify that*
$$\text{SWAP} = \frac{1}{2} \sum_{z,x \in \{0,1\}} Z^z X^x \otimes X^x Z^z.$$

Exercise 5 (Teleportation identity)**.** *The goal of this exercise is to explicitly verify the teleportation identity*

$$|\psi\rangle \otimes |\beta_{00}\rangle = \frac{1}{2} \sum_{z,x \in \{0,1\}} |\beta_{zx}\rangle \otimes X^x Z^z |\psi\rangle. \tag{1}$$

Let $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ *be an arbitrary single-qubit state.*

1. *Substitute* $|\psi\rangle$ *and* $|\beta_{00}\rangle$ *in* $|\psi\rangle \otimes |\beta_{00}\rangle$ *and expand it as a linear combination of the computational basis states. This gives you an expression for the left-hand side of eq. (1).*
2. *Compute* $X^x Z^z |\psi\rangle$ *for all* $x, z \in \{0,1\}$*. Then tensor each of these states with the corresponding Bell state* $|\beta_{zx}\rangle$ *and compute* $|\beta_{zx}\rangle \otimes X^x Z^z |\psi\rangle$*, for each* $x, z \in \{0,1\}$*. Expand each of these four expressions as linear combinations of the computational basis states. This gives you the four terms on the right-hand side of eq. (1).*
3. *Sum the four terms together and verify that both sides of eq. (1) agree.*

Exercise 6 (`Cirq`: Teleportation)**.**